MTH 201 Chapter 3: Infinite Series

3.8 Root Test Worksheet

Determine whether each of the following series converge or diverge. *You may use any theorem from 3.8 or earlier.*

1. $\sum_{k=1}^{\infty} \frac{(-2)^k}{k^k}$

We note that $\sum_{k=1}^{\infty} \frac{(-2)^k}{k^k} = \sum_{k=1}^{\infty} \left(\frac{-2}{k}\right)^k$

We calculate that:

$\rho = \lim_{k\to\infty} \left(\left(\left| \left(\frac{-2}{k} \right) \right| \right)^k \right)^{1/k} = \lim_{k\to\infty} \left(\frac{2}{k} \right) = 0 < 1$. Therefore, by the root test, the series $\sum_{k=1}^{\infty} \frac{(-2)^k}{k^k}$ converges.

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2. $\sum_{k=1}^{\infty} \arctan\left(k^2+1\right)$

We see that $\lim_{k\to\infty} \left(\arctan\left(k^2+1\right) \right) = \frac{\pi}{2} \neq 0$ Therefore, by the test for divergence, the series $\sum_{k=1}^{\infty} \arctan\left(k^2+1\right)$ diverges.

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3. $\sum_{k=1}^{\infty} \frac{1}{k^3+2k+1}$

We note that $\frac{1}{k^3+2k+1} \leq \frac{1}{k^3}$ for $k \geq 1$. We note that both series contain only positive terms. The series $\sum_{k=1}^{\infty} \frac{1}{k^3}$ is a *p*-series with $p = 3 > 1$. Thus $\sum_{k=1}^{\infty} \frac{1}{k^3}$ is convergent. Therefore, by the comparison test, the series $\sum_{k=1}^{\infty} \frac{1}{k^3+2k+1}$ is convergent.

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4. $\sum_{k=1}^{\infty}\left(\frac{k}{k+1}\right)^{k^2}$

The given series contains only positive terms so we compute:

$$\lim_{k\to\infty}\sqrt[k]{\left(\left|\frac{k}{k+1}\right|\right)^{k^2}}=\lim_{k\to\infty}\left(\frac{k}{k+1}\right)^{k}=\lim_{k\to\infty}\frac{1}{\left(\frac{k+1}{k}\right)^{k}}=\lim_{k\to\infty}\frac{1}{\left(1+\frac{1}{k}\right)^{k}}=\frac{1}{e}<1.$$ Therefore, by the root test, the series $\sum_{k=1}^{\infty}\left(\frac{k}{k+1}\right)^{k^2}$ converges.

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5. $\sum_{k=1}^{\infty}\frac{1}{5^k+1}$

We note that $\frac{1}{5^k+1}\le\frac{1}{5^k}$ for $k\ge 1$. We note that both series contain only positive terms. We will investigate the series $\sum_{k=1}^{\infty}\frac{1}{5^k}$. We note that $\rho=\lim_{k\to\infty}\frac{|a_{k+1}|}{|a_k|}=\lim_{k\to\infty}\frac{\left(\frac{1}{5^{k+1}}\right)}{\left(\frac{1}{5^k}\right)}=\lim_{k\to\infty}\frac{1}{5^{k+1}}\cdot\frac{5^k}{1}=\lim_{k\to\infty}\frac{1}{5}=\frac{1}{5}<1$.

Therefore, by the ratio test, the series $\sum_{k=1}^{\infty}\frac{1}{5^k}$ is convergent. Therefore, by the comparison test, the series $\sum_{k=1}^{\infty}\frac{1}{5^k+1}$ is convergent. (Note that the series $\sum_{k=1}^{\infty}\frac{1}{5^k}$ could also be recognized as a convergent geometric series.)

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6. $\sum_{k=1}^{\infty}\left(\frac{1-2k}{k+6}\right)^{k}$

We calculate that:

$\rho=\lim_{k\to\infty}\left(\left(\left|\frac{1-2k}{k+6}\right|\right)^{k}\right)^{1/k}=\lim_{k\to\infty}\left(\frac{2k-1}{k+6}\right)=2>1$. Therefore, by the root test, the series $\sum_{k=1}^{\infty}\left(\frac{1-2k}{k+6}\right)^{k}$ diverges.

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7. $\displaystyle\sum_{k=1}^{\infty}\frac{k^{-1/2}}{2+\sin^2(k)}$

We will progress by using the comparison theorem. We note that $\dfrac{k^{-1/2}}{2+\sin^2(k)} \geq \dfrac{k^{-1/2}}{2+1} = \dfrac{1}{3k^{1/2}}$ for $k \geq 1$.

We see that both series contains only positive terms. The series $\displaystyle\sum_{k=1}^{\infty}\frac{1}{3k^{1/2}} = \frac{1}{3}\sum_{k=1}^{\infty}\frac{1}{k^{1/2}}$ is a *p*-series with $p = \dfrac{1}{2} < 1$. Thus $\displaystyle\sum_{k=1}^{\infty}\frac{1}{3k^{1/2}}$ is divergent. Therefore, by the comparison theorem, since $\displaystyle\sum_{k=1}^{\infty}\frac{1}{3k^{1/2}}$ diverges the series $\displaystyle\sum_{k=1}^{\infty}\frac{k^{-1/2}}{2+\sin^2(k)}$ diverges.

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8. $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k^2-5}{2k^2+1}\right)^k$

We want to use the Root Test, but we note that simplifying the absolute value of $\dfrac{k^2-5}{2k^2+1}$ is tricky because it is sometimes negative and sometimes positive. We note that when $k \leq 2$ the expression is negative and is positive for all $k \geq 3$. Therefore, we first rewrite the series as $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k^2-5}{2k^2+1}\right)^k = \left(\frac{-4}{3}\right) + \left(\frac{-1}{9}\right)^2 + \sum_{k=3}^{\infty}\left(\frac{k^2-5}{2k^2+1}\right)^k$. We then use the Root test on the new series.

Note that $\left|\dfrac{k^2-5}{2k^2+1}\right| = \dfrac{k^2-5}{2k^2+1}$ for $k \geq 3$. We then calculate

$$\lim_{k\to\infty}\left(\left|\frac{k^2-5}{2k^2+1}\right|^k\right)^{1/k} = \lim_{k\to\infty}\frac{k^2-5}{2k^2+1} = \lim_{k\to\infty}\frac{k^2-5}{2k^2+1}\cdot\frac{1/k^2}{1/k^2} = \lim_{k\to\infty}\frac{1-5/k^2}{2+1/k^2} = \frac{1}{2} < 0$$

Therefore, by the Root Test, the series $\displaystyle\sum_{k=3}^{\infty}\left(\frac{k^2-5}{2k^2+1}\right)^k$ converges. Then, since the series $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k^2-5}{2k^2+1}\right)^k$ is just the series $\displaystyle\sum_{k=3}^{\infty}\left(\frac{k^2-5}{2k^2+1}\right)^k$ with two extra terms added, this series also converges.

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